

Revisiting Implicit Curriculum in RLVR

Reproduction and local analysis of synthetic state tracking

Yankai Ding

Abstract

A flat terminal-success curve can conceal changes in how a model solves a task. We revisit the synthetic state-tracking experiment of Huang et al. [1] through an independent implementation. Mixed training improves long-task performance in the main run, but the reported learning relay remains incomplete across three seeds and an added five-level mixture. We designed local measurements of progress within trajectories and through reward gradients. In the main run, attention allocation and local execution improve before terminal success rises appreciably. Reward-gradient norms peak in length order and then decline unevenly. L15 directions help L45 more often late in training, even after the L15 gradient has weakened. Stable operation-selection accuracy, sharpening attention, and polarized prompt outcomes characterize the resulting partial policy. These findings reveal internal progress and remaining opportunities for cross-length improvement within a persistent plateau; why training enters this regime remains unresolved.

1 Learning within a fixed task mixture

Huang et al. [1] show that a fixed mixture of task lengths can generate a learning relay from short to long tasks. This raises two related questions: whether the relay appears in an independent implementation, and what changes locally as long-task performance develops. Section 2 compares terminal learning curves. Sections 3 and 4 examine the process within trajectories and through reward gradients, respectively.

Each prompt contains an initial state and L distinct additions modulo 96. Thus, L counts the operations in the intended solution. A fixed hidden permutation links query identifiers to operation positions. At each step, attention forms a weighted operation mixture; a frozen MLP combines it with the generated current state to predict the next state. Reward tests the final state:

$$s_L^* = (s_0 + \sum_{k=1}^L g_k) \bmod 96, \quad r = \mathbf{1}\{s_L = s_L^*\}.$$

Here s_L^* is the intended final state and s_L is the generated state. RL trains only the 64×64 attention matrix Q , initialized to zero. The shared frozen MLP solves every atomic state-operation pair. A uniform terminal guess succeeds with probability $1/96$, or about 1.04%. Longer tasks require more successive operations before this terminal feedback arrives.

Condition	Training lengths	Performance evaluation
Fixed	45 only	$L = 45$
Mixed	Uniform per sample: 5, 15, 45	$L = 5, 15, 45$

Each run uses 40,000 updates and batches of 512 trajectories. We evaluate 15,360 prompts per length every 50 updates. The main analysis follows the original run, seed 101. Two additional seeds test how the outcome varies; a separate five-level mixture retains L45 while adding intermediate lengths. Implementation details and these comparisons appear in the appendices.

2 Reproducing the terminal learning curves

The original $R = 3$ experiment reports a smooth learning relay from L5 through L15 to L45 [1, §7.1, Figure 4a]. We reconstruct the mixed L5/15/45 condition and compare it with L45-only training. Both conditions share the transition network, initialization, optimizer, and update count. Figure 1 tests the long-task benefit of mixing and the progression across lengths.

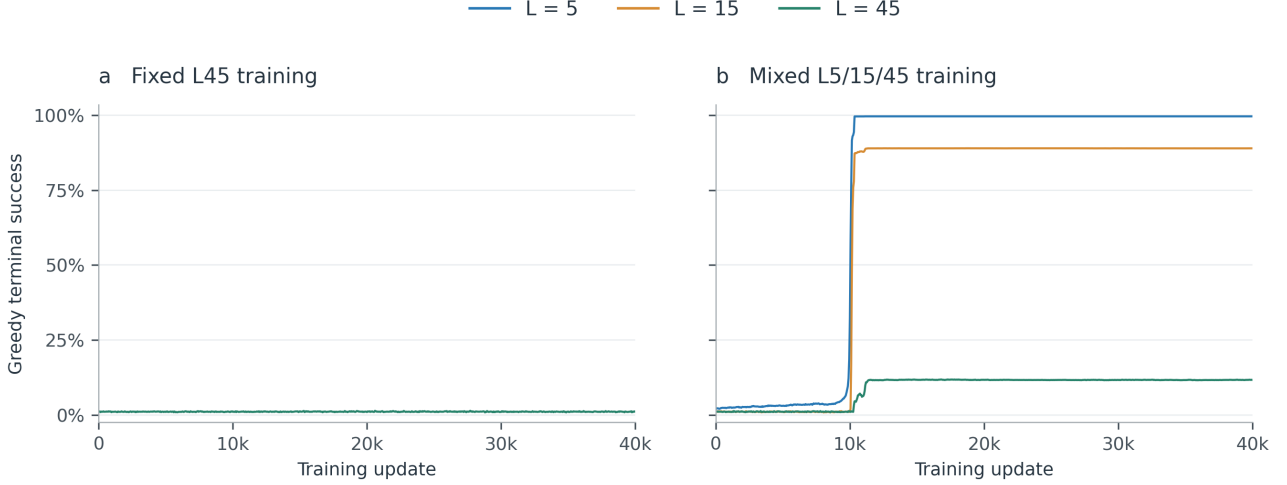


Figure 1. Terminal learning under fixed and mixed training. Left: L45-only training. Right: uniform mixed training on L5, L15, and L45. Colors denote evaluation lengths. Both panels show greedy terminal success on identical axes, retaining all 801 evaluations.

A rapid but incomplete transition. Fixed L45 training remains near chance. In the mixture, L5 and L15 first exceed 50% terminal success at updates 10,050 and 10,200. L45 improves to 11.65%, compared with 1.03% in the fixed control. This run reproduces the long-task benefit of mixed training, while the reported smooth relay remains incomplete.

Condition	Greedy terminal	Sampled terminal
Fixed, L = 45	1.03 ± 0.08	0.94 ± 0.08
Mixed, L = 5	99.67 ± 0.05	99.45 ± 0.06
Mixed, L = 15	88.99 ± 0.25	88.35 ± 0.26
Mixed, L = 45	11.65 ± 0.26	11.46 ± 0.26

Table 1. Final terminal success at update 40,000. Percentages with one standard error over 15,360 prompts. Greedy and sampled decoding yield similar outcomes.

Variation across runs. With the same implementation and frozen head, seeds 102 and 103 reach 66.17% and 1.01% mixed L45 success. None of the three seeds completes the relay (Appendix C). The additional L5/10/20/40/45 mixture reaches another partial solution (Appendix E).

What the plateau leaves open. A terminal learning curve alone does not show whether operation selection has improved, how reliably individual operations are executed, or whether learning on one length can locally help another. We examine these questions in the original seed-101 run: first within its generated trajectories, then through its reward gradients.

3 Internal progress before terminal success

We designed two measurements to inspect the L45 computation. *Correct attention mass* is the weight assigned to the intended operation, averaged over all steps and prompts. *Local correct-transition probability* is the probability of correctly applying that operation from the model's actual sampled current state, which may differ from the canonical state; it is also averaged over all steps and prompts. Sampled terminal success provides the outcome reference.

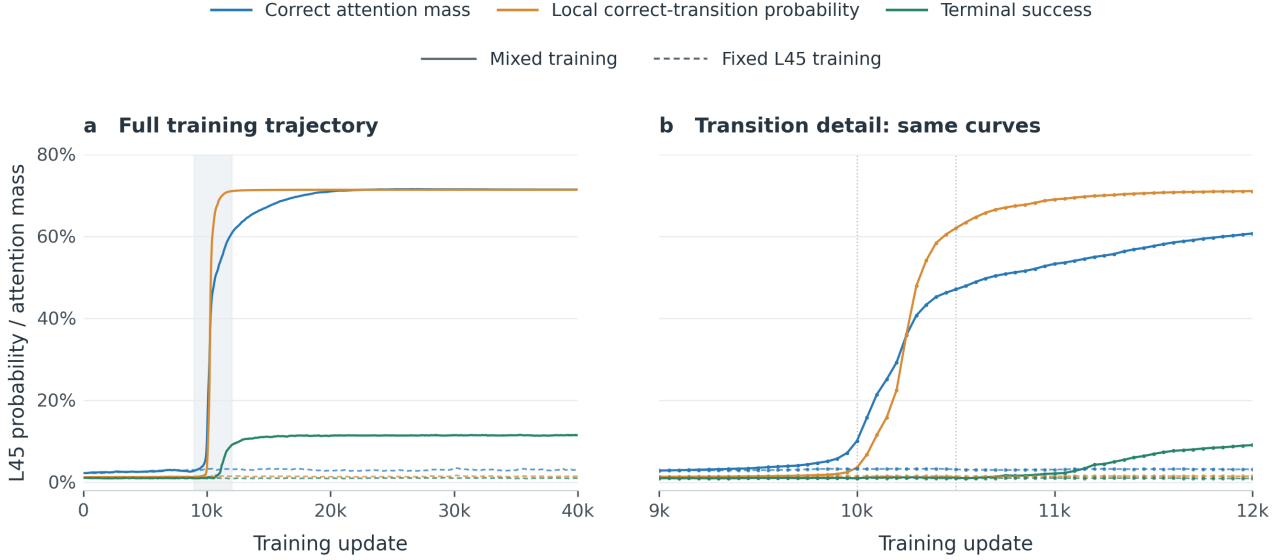


Figure 2. Two local measurements along the learning process. Both panels show the same L45 quantities and training conditions. Left: all 801 evaluations, with the transition window shaded. Right: the same curves over updates 9,000-12,000, retaining all 61 measurements. Colors identify metrics; solid and dashed lines identify mixed and fixed training. The window was selected after the run; guides mark 10,000 and 10,500. No smoothing is applied.

At update 10,000, mixed L45 sampled success is still 0.91%, while correct attention mass has increased from 2.22% at initialization to 10.08%. By update 10,500, attention mass reaches 47.05% and local correct-transition probability reaches 61.99%, yet sampled terminal success is only 1.11%. Internal changes precede a substantial rise in terminal success.

L45 measurement	Mixed training	Fixed training
Correct attention mass	47.05	3.11
Local correct-transition probability	61.99	1.42
Sampled terminal success	1.11	0.85

Table 2. L45 at update 10,500. All entries are percentages, evaluated on the same 15,360 prompts in both conditions.

Together, the trajectories show early internal progress followed by lasting but incomplete gains. Under mixed training, attention mass and local transition probability settle near 71.4%, while sampled terminal success remains near 11.5%. Fixed training ends at 3.00% attention mass and 1.40% local transition probability. The improvement therefore extends to both operation selection and local execution, but stops short of reliable long-task completion. Averaging local correctness over steps and prompts does not determine full-path success: it leaves unresolved how errors are distributed across complete trajectories.

4 Comparisons Based on Reward Gradients

4.1 Reward gradients under mixed and fixed training

At each checkpoint, we measure reward gradients for $L = 5, 15$, and 45 offline, using three fixed 16-prompt batches per length. Measurements occur every 100 training updates and leave Q unchanged. Integrating over generated trajectories gives terminal success probability $P_L(Q; x)$ and its length-normalized gradient:

$$g_L(Q) = \nabla_Q \left[\frac{1}{nL} \sum_{i=1}^n P_L(Q; x_i) \right], \quad n = 16.$$

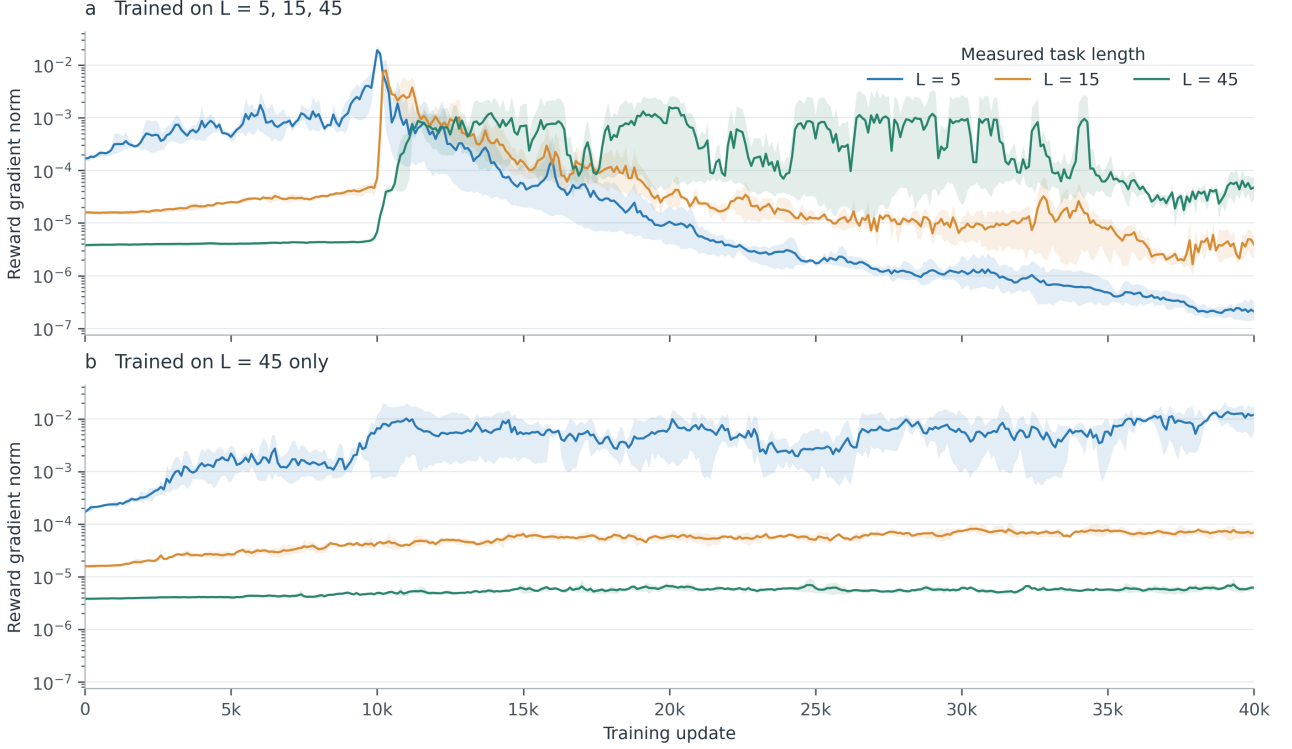


Figure 3. Three reward objectives along each training trajectory. Panel titles specify training lengths; colors specify measurement lengths. Curves within a panel use the same model checkpoints. Lines show mean batch norms; shading spans the three batches. Both panels use identical prompts and axes, $1/L$ normalization, and no smoothing.

Ordered growth and unequal decay. Under mixed training, the $L5$, $L15$, and $L45$ gradients rise in sequence and reach sampled maxima at updates 10,000, 10,300, and 20,000. The temporal resolution is 100 updates, so the exact peak steps may differ. $L5$ and $L15$ then decline with fluctuations; $L45$ remains elevated longer before shifting downward around 35,000. From 17,600 onward, $\|g_{45}\| > \|g_{15}\| > \|g_5\|$ holds at every checkpoint, reversing the initial ordering.

The fixed control. With $L45$ -only training, the measured $L5$ gradient grows to 12.9 times its initial norm by update 5,000 and 39.2 times by 10,000. Yet $\|g_5\| > \|g_{15}\| > \|g_{45}\|$ holds throughout, and the $L45$ norm never exceeds 1.87 times its initial value. Thus, growing short-task sensitivity alone does not bring the long-task transition.

Interpretation. The mixture stays fixed while the relative gradient norms reverse. Their decline as short-task success rises is consistent with reward saturation. This is a local signature of an implicit curriculum: learning changes the relative reward sensitivity of the task lengths.

The fixed run develops strong short-task gradients but never uses short-task rewards. The mixed run shows both short-task saturation and greater long-task sensitivity. Gradient norms describe the strength of these signals; Section 4.2 asks whether their directions can still help $L45$ as the short-task signals weaken.

4.2 Easy-hard gradient alignment and fine-tuning consistency

At checkpoints of the mixed run, we compare L5-L45 with L15-L45. Gradient alignment measures whether the rewards favor similar parameter changes. A temporary step of fixed size along each easier-task gradient tests *fine-tuning consistency*: whether that direction also improves L45. Three fixed task sets (A-C), each with 16 prompts per length, supply the source gradients and target rewards.

For source length $l = 5$ or 15 , we compute the gradient cosine c and move Q by $\varepsilon = 0.01$ along the unit source gradient. The L45 response Δ measures compatible improvement when positive and interference when negative.

$$c_{l,45} = \frac{\langle g_l, g_{45} \rangle}{\|g_l\| \|g_{45}\|}, \quad Q^+ = Q + \varepsilon \frac{g_l}{\|g_l\|},$$

$$\Delta_{l \rightarrow 45} = \bar{P}_{45}(Q^+) - \bar{P}_{45}(Q).$$

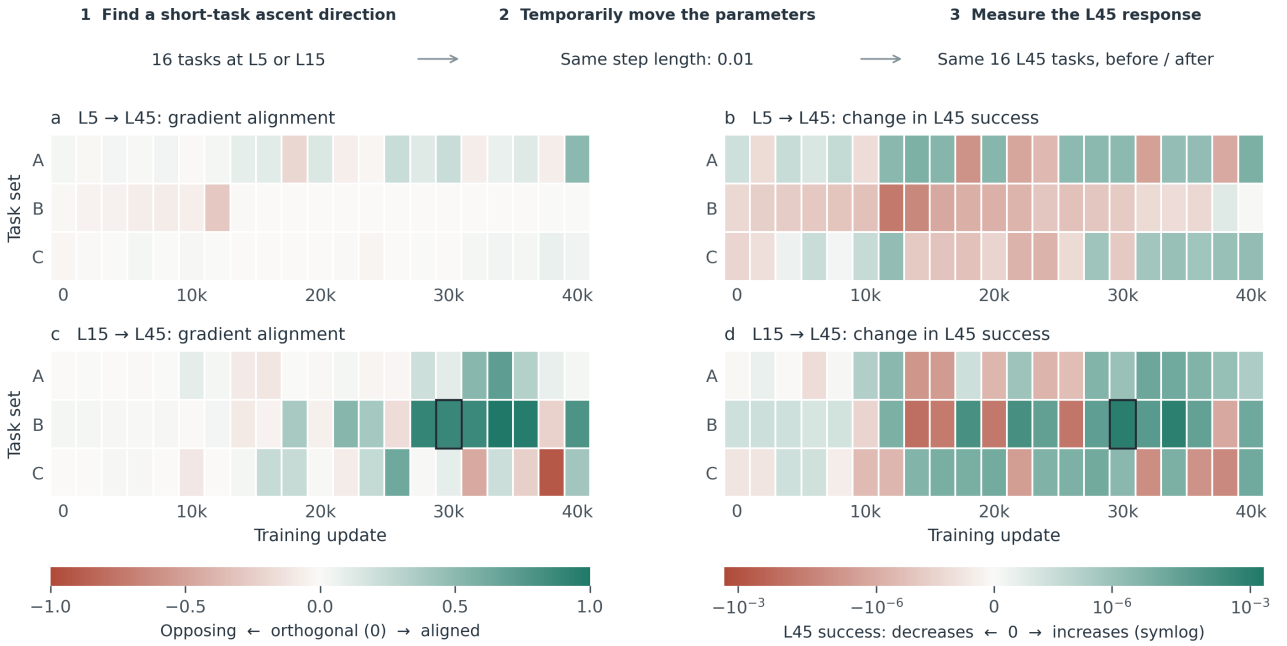


Figure 4. Direction agreement and measured long-task response. Left: gradient cosine; green is aligned, red opposing, and white nearly orthogonal. Right: the fine-tuning response Δ in L45 success probability; green improves success, red reduces it. Each cell is one of three task sets at a checkpoint, every 2,000 updates. L5 and L15 share the L45 target set. Left colors are linear; right colors use a symmetric log scale, linear within $\pm 10^{-8}$. All measurements are shown. Outlines mark the example below.

The largest measured L15 gain occurs for set B at update 30,000 in panels (c,d). The outlined left cell has a cosine of 0.89: the L15 and L45 reward gradients point in similar directions. The matching right cell records $\Delta = 1.11 \times 10^{-3}$, raising L45 success from 8.00% to 8.11%. The paired cells illustrate fine-tuning consistency: an L15 ascent step improves the L45 reward.

Weak early coupling. Through update 10,000, all absolute gradient cosines are below 0.105 and $|\Delta| < 3.35 \times 10^{-7}$. Neither source consistently benefits L45 in this early window.

More frequent late improvement from L15. Over updates 12,000-20,000, L15 steps improve L45 in 8 of 15 measured cells. Over updates 22,000-40,000, they help in 22 of 30 cells, compared with 14 of 30 for L5. L15-L45 alignment and improvement become more consistently positive late in training, with substantial variation across task sets and effect sizes. These counts describe repeated measurements of the same three sets.

Interpretation. L15 directions more often help L45 late in training, while the L15 reward-gradient norm has sharply declined. Fixing the parameter-step size isolates directional opportunity from source strength. Together, Figures 3 and 4 show that useful cross-length directions can persist after the easy-task signal weakens. The next question is how much these remaining directions contribute to actual optimizer updates and sustained learning.

5 Computing and validating the local measurements

Expected reward over generated trajectories

The gradient measurements require terminal probabilities even when successful rollouts are rare. For a fixed prompt, the 96-state task permits a finite-state computation that sums over every possible generated trajectory. At step k , the frozen MLP defines a transition matrix $T_k(Q)$, whose rows give next-state probabilities. Starting from the initial state, we propagate this distribution and read out the probability of the intended endpoint:

$$p_0 = e_{s_0}^T, \quad p_k = p_{k-1} T_k, \quad P_L = p_L e_{s_L^*}.$$

A backward recursion gives the derivative with respect to Q . Centering its backward values reduces numerical cancellation: adding a constant to those values contributes zero derivative through a transition row that sums to one. Computation uses float64. This integrates trajectory randomness exactly for the fixed prompts; averaging across prompts still uses the three sampled task sets.

From gradient alignment to a measured response

For the unit source-gradient step in Section 4.2, first-order expansion connects the predicted L45 reward change to its gradient norm and alignment with the source:

$$\Delta_{l \rightarrow 45} \simeq 45 \varepsilon \|g_{45}\| c_{l,45}.$$

The factor 45 reverses the length normalization. Alignment determines the predicted sign; the target norm and step size determine its magnitude. All 126 responses in Figure 4 have the predicted sign, validating this local derivative interpretation. Positive and negative perturbations at two step sizes also check derivative accuracy (Appendix B). Actual training updates additionally depend on source gradient magnitudes, Adam, the baseline, and entropy regularization.

6 Conclusion

Mixed training can substantially improve long-task performance in this independent implementation, but none of the three runs completes the original learning relay. The five-level mixture also reaches partial success. Our local measurements examine the progress and plateaus within these outcomes.

Attention shifts toward the intended operations and local execution improves before terminal success rises appreciably. Reward-gradient norms peak in length order and decline unevenly. Later in the main run, L15 directions help L45 more often even as the L15 signal weakens. Local opportunities for improvement therefore coexist with a persistent performance plateau.

In seed 101, operation-selection accuracy stops improving while attention sharpens, probe outcomes polarize, and short-task reward gradients shrink (Appendix D). Together, these observations support saturation of a partial policy despite continued parameter updates. Other seeds reach different plateau levels, including a run with substantial remaining short-task gradients. Why these regimes emerge remains unresolved; the learned transition network and the contributions of each length to actual optimizer updates are the next objects of analysis.

AI use

OpenAI Codex assisted extensively with experimental design, code, execution, analysis, figures, and writing. The author directed the study and is responsible for the final report.

Reference

[1] Yu Huang, Zixin Wen, Yuejie Chi, Yuting Wei, Aarti Singh, Yingbin Liang, and Yuxin Chen. *On the Emergence of Implicit Curriculum in RLVR Learning Dynamics*. ICML, PMLR 306:47891-47938, 2026. Conference paper. Experimental exposition: arXiv:2602.14872v3, §7.1, Figures 3-4.

Appendix A Implementation and experimental controls

We independently implemented the task and attention architecture described in [1]; author code and the trained transition head were not obtained. The table specifies the reconstruction and its implementation choices. The same pretrained head and settings are used throughout the formal runs. We follow the terminal-state reward formula and also measure full-path success, which requires every canonical intermediate state.

Component	Implementation
Task	Additions modulo 96, distinct operations, uniform initial states, and a fixed identifier permutation [1, §§3.1, 7.1]. We use 64 identifiers and task seed 2026.
Attention and head	Class-specific ReLU architecture with operation and state inputs scaled by 1/2 [1, Eqs. (3)-(4)]. We use 384 units per output class and a 64×64 Q initialized to zero.
Head pretraining	6,000 Adam updates; learning rate 0.003; batch 1,024; seed 17; Gaussian initialization with SD 0.05. Leaky ReLU slope 0.01 during supervision; ReLU during validation and RL.
RL optimizer	Adam at constant learning rate 0.01; betas (0.9, 0.999); epsilon 10^{-8} ; no weight decay.
RL objective	Length-normalized REINFORCE; entropy coefficient magnitude 0.001 [1, §§3.4, 7.1]. In our loss, entropy is averaged over steps and prompts, then subtracted.
Reward baseline	EMA momentum 0.95 [1, §7.1]. We use a global baseline, initially zero. Each update uses the previous baseline, then updates it with the current batch reward.
Training	Batch 512; 40,000 updates; fixed L45 or uniform per-trajectory L5/15/45 mixture. Run seeds 101, 102, and 103. The added five-level mixture uses seed 101.
Evaluation	15,360 prompts per length every 50 updates, including initialization. Greedy and sampled rollouts share prompts; evaluation has an RNG separate from training.
Local measurements	Three fixed sets of 16 prompts per length. Reward gradients every 100 updates; source-to-L45 perturbations every 2,000 updates, with signed steps of 0.01 and 0.005.
Execution	Rented instances with one NVIDIA GeForce RTX 4090; Python 3.12.14 and PyTorch 2.11.0+cu128.

Execution capacity. The frozen head correctly predicts all 9,216 atomic state-operation pairs; the minimum and mean correct probabilities are 0.993072 and 0.999507. In a separate oracle-attention check, we supply the intended one-hot operation at every step of 8,192 new L45 prompts. The mean probability of a fully correct path is 97.81%. The head can therefore support long correct trajectories when the intended operations are selected.

Matched exposure. Every run processes 20,480,000 trajectories. Within each seed, the fixed and mixed conditions share initialization, head, task permutation, optimizer, and evaluation prompts. Matching updates and trajectories leaves different totals of executed operations.

Fixed-control measurements. The original seed-101 collection omitted fixed-group reward gradients. A rerun with the same seed and configuration reproduced every evaluation metric at all 801 checkpoints. Its gradients supply the fixed panel of Figure 3. The five-level comparison adds measurement lengths using this rerun's saved Q states, without further fixed-group training.

Evidence retained. Numeric archives include per-update training records, evaluations, local measurements, configurations, and source hashes. Seeds 102 and 103 retain both conditions' Q snapshots, recovery checkpoints, and selected actual updates; the five-level run retains the same records for its mixed condition. The seed-101 fixed rerun retains 801 Q snapshots. All plots regenerate from local numeric data, with provenance in the report manifest.

Appendix B Training fluctuations and numerical accuracy

Per-update rewards and parameter changes continue to fluctuate after evaluation success plateaus. Figure 5 shows this ongoing optimization throughout the 40,000-update run.

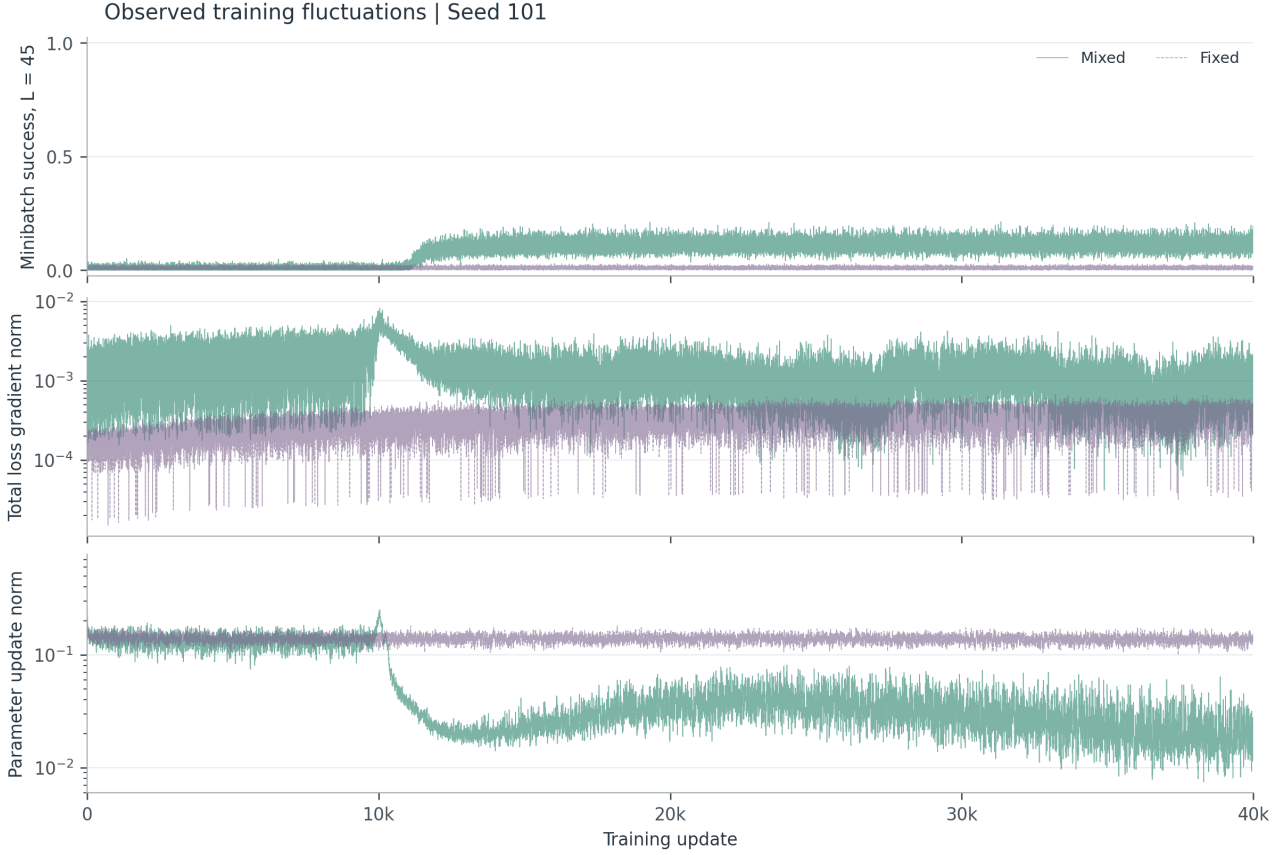


Figure 5. Training activity during partial learning. Top: L45 minibatch success; middle: total loss-gradient norm, including baseline and entropy terms; bottom: Adam update norm. All 40,000 updates are shown.

Derivative check. We validate the expected-reward calculation against trajectory enumeration on small tasks, direct differentiation, and independent rollouts. For each source direction $u_l = g_l / \|g_l\|$, a central difference provides a further check at the measured checkpoints:

$$D_\varepsilon = \frac{\bar{P}_{45}(Q + \varepsilon u_l) - \bar{P}_{45}(Q - \varepsilon u_l)}{2\varepsilon}, \quad d = 45 g_{45}^T u_l.$$

Of the 126 source-target cells, 123 agree within 1% at $\varepsilon = 0.01$ and 125 at $\varepsilon = 0.005$. Table 3 gives the three exceptions at the larger step. Their response signs agree with the directional predictions at both sizes, as do those of all other cells.

Update	Source $\rightarrow$ target	Set	Gap at $\varepsilon = 0.01$	Gap at $\varepsilon = 0.005$
8,000	L5 $\rightarrow$ L45	C	1.54%	1.37%
12,000	L15 $\rightarrow$ L45	C	1.18%	0.17%
38,000	L15 $\rightarrow$ L45	A	1.25%	0.90%

Table 3. Relative derivative error. Smaller steps reduce the error; all three cells preserve the response sign.

Relative error is $|D_\varepsilon - d| / \max(|d|, f)$, with cancellation floor $f = 64 u_{\text{mach}} \max(P_+, P_-, P_0) / \varepsilon$; all derivatives exceed this floor. The 1% tolerance tests derivative accuracy. Whether an easier-task step benefits L45 is determined by the signed reward change in Section 4.2. Reusing the prompt sets makes successive checkpoints dependent measurements.

Appendix C Partial solutions across training seeds

To examine how much the original outcome depends on training randomness, seeds 102 and 103 repeat both conditions and the complete measurement schedule. The code, hyperparameters, frozen head, and identifier permutation stay fixed. Figure 6 compares terminal outcomes and reward-gradient dynamics across all three runs.

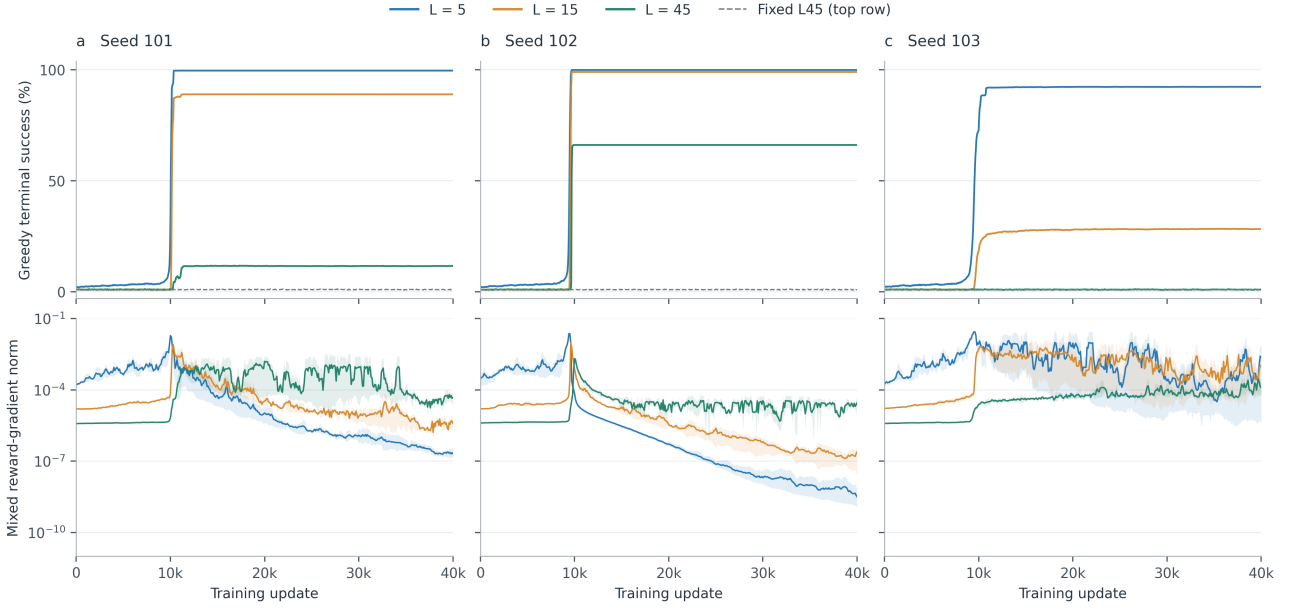


Figure 6. Different outcomes and gradient regimes under one implementation. Each column is a seed. Top: mixed terminal success by length and the dashed fixed-L45 control. Bottom: mean reward-gradient norms, with shading across three probe sets. Axes are shared within rows; all 801 evaluations and 401 gradient checkpoints are retained.

Seed	Metric	Mixed L5	Mixed L15	Mixed L45	Fixed L45
101	Terminal	99.67	88.99	11.65	1.03
101	Full path	91.40	41.11	0.00	0.00
102	Terminal	99.96	99.05	66.17	1.00
102	Full path	97.15	72.58	1.85	0.00
103	Terminal	92.37	28.27	1.01	0.98
103	Full path	66.54	4.02	0.00	0.00

Table 4. Final terminal and full-path success. Percentages on 15,360 prompts per length. Full-path success requires every canonical intermediate state, as well as the correct endpoint.

Different plateau levels. Seed 102 reaches 66.17% L45 terminal success, yet only 1.85% full-path success. Seed 103 remains near chance at L45. Incomplete learning thus recurs across the three seeds, with markedly different amounts of terminal and trajectory-level progress.

Different local regimes. Seed 102 repeats the ordered gradient peaks, at updates 9,400, 9,600, and 10,000, followed by strong short-task decay. Seed 103 instead retains substantial L5 and L15 gradients. Its L45 attention mass and local transition probability reach 41.75% and 41.68%, despite chance-level terminal success. A performance plateau does not identify a single underlying state of learning.

Choice of the main case. Seed 101 was the original run. Before the two additional runs, we specified that a successful replacement must sustain at least 99% greedy terminal success at every training length throughout updates 39,000–40,000. Neither qualifies, so the main analysis retains seed 101 and the comparison reports all three outcomes.

Appendix D Operation selection in the L15 plateau

Peak attention-hit rate, also measured in [1, Figure 3b], counts steps at which the largest attention weight selects the intended operation. We compare it with the attention mass from Section 3 to distinguish a gain in selection accuracy from increasing weight on an existing selection. Figure 7 follows this discrete accuracy under fixed and mixed training.

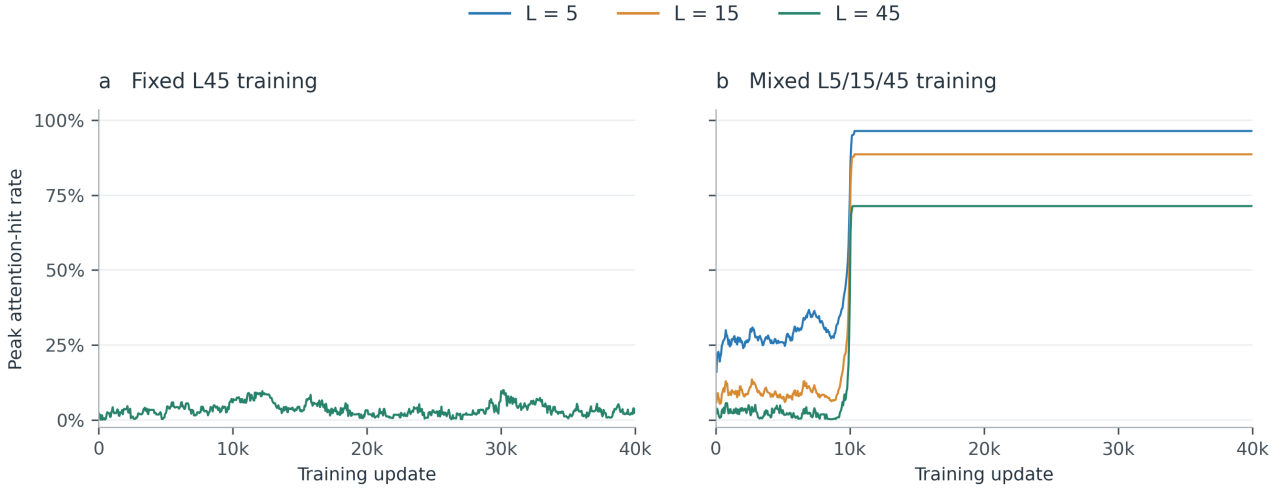


Figure 7. Accuracy of operation selection. The left panel follows the fixed-L45 run; the right follows mixed L5/15/45 training. Colors identify evaluation lengths. Both panels use the same 801-checkpoint grid as Figure 1. The mixed-condition measurement extends the source paper’s fixed-length attention comparison.

Condition	Attention hit	Full path
Fixed, L = 45	2.74	0.00
Mixed, L = 5	96.45	91.40
Mixed, L = 15	88.66	41.11
Mixed, L = 45	71.38	0.00

Table 5. Operation selection and complete trajectories at update 40,000. Percentages on 15,360 prompts. Attention hit averages position selection over steps; full-path success requires every canonical state along the trajectory.

Sharpening within a partial solution. L15 full-path success stays at 41.11% from update 10,400. Between updates 12,000 and 40,000, correct attention mass rises from 83.08% to 88.67%, while aggregate attention-hit rate remains 88.66%. The allocation sharpens without a net gain in correct position selection. Meanwhile, the L15 reward-gradient norm falls about 2,054-fold from its peak, even as parameter updates continue.

Polarized prompt outcomes. At the final checkpoint, 42 of the 48 fixed L15 probe prompts average 99.26% terminal success probability; the remaining six average 0.029%. These groups are defined retrospectively from the probe set. Their separation, together with the diminishing gradient, supports saturation of an incomplete policy as an explanation for how the plateau persists.

What terminal reward preserves. Among rewarded L15 training trajectories during updates 20,001–40,000, 54.38% contain an incorrect intermediate state. Such trajectories receive the same terminal reward as fully correct paths. The objective therefore allows high terminal success without faithful execution of the entire operation sequence.

Appendix E A five-level mixture retaining L45

We add a mixture of L5, L10, L20, L40, and L45 to examine denser intermediate coverage while keeping the same longest task. Lengths are sampled uniformly from the first update. The schedule doubles from 5 until the cap at 45; L15 is measured but never trained. This is a five-level experiment. The three-level L5/10/20 design remains untested.

Seed 101, the frozen head, optimizer, batch size, and update horizon match the main run. Changing the mixture reduces the L45 sampling share from one third to one fifth and increases mean task length from 21.67 to 24. The comparison measures this joint change in task composition and exposure.

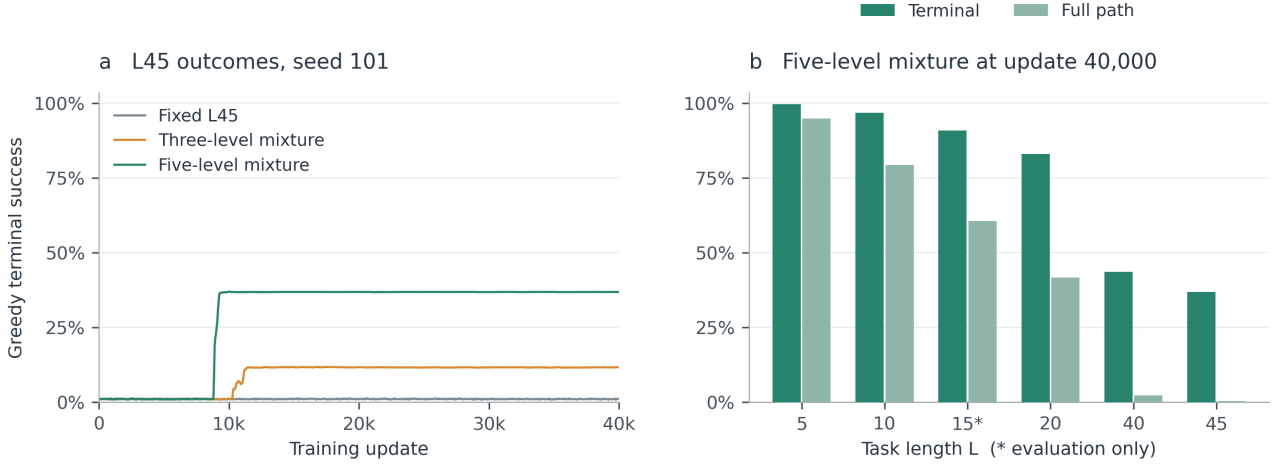


Figure 8. An improved but incomplete five-level outcome. (a) L45 terminal success for fixed training and the three- and five-level mixtures, all with seed 101. (b) Terminal and full-path success of the five-level run at update 40,000. Each rate uses 15,360 prompts; the asterisk marks untrained L15. All 801 time points are retained.

A different plateau. L45 terminal success reaches 36.88%, compared with 11.65% in the three-level seed-101 run and 1.03% under fixed training. L45 full-path success is 0.39%. The terminal rate stays between 36.82% and 36.88% over updates 20,000-40,000. The added lengths improve this seed's outcome, while the 66.17% result of three-level seed 102 remains higher.

Ordered sensitivity, uneven coupling. Mean reward-gradient norms peak at updates 8,600, 8,700, 8,800, 9,000, 9,200, and 9,200 for L5, L10, L15, L20, L40, and L45. The two longest lengths share a peak on the 100-update grid. At the final checkpoint, gradient norms increase with length. In the local perturbations at $\epsilon = 0.01$, L40 directions help L45 in 57 of 63 cells, whereas L20 directions help it in 25. All 315 changes have the predicted sign. Local assistance is present even within this persistent plateau.

Matched measurements. All six lengths use three fixed sets of 16 prompts and 100-update gradient measurements. Every 2,000 updates, we step along each shorter-length reward gradient at $\epsilon = 0.01$ and 0.005 and measure the L45 response. Fixed-group gradients are computed from its 801 saved Q states; all 3,609 measurements at the original lengths match the earlier values exactly.